

Task 2 - Deep Learning Image Classification

InternSpark Internship Submission

Project: CIFAR-10 Image Classification using PyTorch and Pretrained ResNet18

1. Project Overview

This project implements a small deep learning image-classification model using PyTorch. A pretrained ResNet18 model is used with transfer learning to classify images from the CIFAR-10 dataset. Data augmentation is applied to the training images, and the trained model is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix.

2. Technologies and Environment

Programming language: Python

Framework: PyTorch 2.13.0+cpu

Computer device: CPU execution (CUDA unavailable in the notebook)

Libraries: Torchvision, Scikit-learn, Matplotlib

3. Dataset and Data Augmentation

The CIFAR-10 dataset contains 10 image classes. The notebook downloads the complete CIFAR-10 dataset locally, then uses a smaller subset for CPU-friendly training and evaluation.

Item	Value
Training subset	5,000 images
Testing subset	1,000 images
Training batch size	64
Training augmentation	Random horizontal flip; random rotation (10°)
Image preprocessing	Resize to 224×224; tensor conversion; normalization

4. Model and Training

The notebook uses a pretrained ResNet18 model and replaces its final fully connected layer with a 10-class output layer for CIFAR-10. Cross-Entropy Loss is used as the loss function and Adam is used as the optimizer with a learning rate of 0.001. The model is trained for 2 epochs to keep CPU training time manageable.

Metric	Result
Epoch 1 training accuracy	65.78%
Epoch 1 test accuracy	68.00%

Epoch 2 training accuracy	77.66%
Epoch 2 test accuracy	73.30%
Macro precision	0.74
Macro recall	0.73
Macro F1-score	0.73

5. Evaluation

The final evaluation was performed on 1,000 test images. The classification report reports an overall accuracy of 73%, with macro precision of 0.74, macro recall of 0.73, and macro F1-score of 0.73. A confusion matrix was also generated to visualize predictions across the ten CIFAR-10 classes.

The saved model file is **resnet18_cifar10.pth**.

6. Training Curves and Confusion Matrix

The notebook includes plots for training/test loss and training/test accuracy, as well as a CIFAR-10 confusion matrix. These visualizations are included in the submitted notebook.

Note: The notebook itself contains the generated plots and their recorded outputs.

7. Inference

An inference function named **predict_image()** is included in the notebook. It loads an image, applies the test transformation, runs the trained model, and prints the predicted CIFAR-10 class.

Example command:

```
predict_image("path/to/your/image.jpg")
```

8. Files Included

The Task 2 GitHub folder should contain:

- Task2_Deep_Learning.ipynb
- README.md
- resnet18_cifar10.pth

The downloaded CIFAR-10 **data/** folder does not need to be uploaded to GitHub.

9. Environment Setup and Commands

Install the required packages:

```
pip install torch torchvision scikit-learn matplotlib
```

Start Jupyter Notebook:

```
jupyter notebook
```

Open **Task2_Deep_Learning.ipynb** and run the notebook cells in order.

10. Conclusion

A pretrained ResNet18 model was successfully fine-tuned for CIFAR-10 image classification using PyTorch. The project demonstrates transfer learning, data augmentation, model training, evaluation metrics, visualization, model saving, and an inference function. The model achieved 73.30% test accuracy on the 1,000-image evaluation subset.